```c
#include <stdio.h>
#include <string.h>

// ---------- Product Data Module ----------
struct Product {
    char name[50];
    float price;
    float rating;
};

// ---------- Sorting Module (Simple sort) ----------
void sortProducts(struct Product p[], int n, int choice)
{
    struct Product temp;

    for (int i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            // Sort by Price
            if (choice == 1 && p[i].price > p[j].price)
            {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }

            // Sort by Rating
            else if (choice == 2 && p[i].rating < p[j].rating)
            {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }

            // Sort by Name
            else if (choice == 3 && strcmp(p[i].name, p[j].name) > 0)
            {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }
}

// ---------- Display Module ----------
void displayProducts(struct Product p[], int n)
```

```
{
    printf("\n Product List:\n");

    printf("-----------------------------------------------\n");

    for (int i = 0; i < n; i++)
    {
        printf("%d. %-15s Rs.%-8.2f Rating: %.1f\n",
               i + 1, p[i].name, p[i].price, p[i].rating);
    }
}

// ---------- Main UI (User Selection Module) ----------
int main()
{
    int n, choice;

    printf("======= E-Commerce Product System =======\n");
    printf("Enter number of products: ");
    scanf("%d", &n);

    struct Product p[n];

    // ---------- Input Module ----------
    for (int i = 0; i < n; i++)
    {
        printf("\nEnter product %d name: ", i + 1);
        scanf("%49s", p[i].name);

        printf("Enter price: ");
        scanf("%f", &p[i].price);

        printf("Enter rating (1-5): ");
        scanf("%f", &p[i].rating);
    }

    // ---------- Menu UI ----------
    while (1)
    {
        printf("\n======= MENU =======\n");
        printf("1. Sort by Price (Low -> High)\n");
        printf("2. Sort by Rating (High -> Low)\n");
        printf("3. Sort by Name (A -> Z)\n");
        printf("4. Display Products\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```
        switch (choice)
        {
            case 1:
            case 2:
            case 3:
                sortProducts(p, n, choice);
                printf("Sorted Successfully!\n");
                break;

            case 4:
                displayProducts(p, n);
                break;

            case 5:
                printf("Exiting...\n");
                return 0;

            default:
                printf("Invalid choice!\n");
        }
    }

    return 0;
}
```